13 In reaction 1, hydrogen reacts with gaseous bromine to form hydrogen bromide.

$$H_2(g) + Br_2(g) \rightarrow 2HBr(g)$$

In reaction 2, hydrogen reacts with gaseous iodine to form hydrogen iodide.

$$H_2(g) + I_2(g) \rightarrow 2HI(g)$$

For reaction 1, the rate equation is

$$\text{rate} = \frac{k_1[H_2][Br_2]^{1.5}}{[Br_2] + k_2[HBr]}$$

For reaction 2, the rate equation is

$$\text{rate} = k[H_2][I_2]$$

What is correct about these reactions?

1 For the hydrogen and bromine reaction, the formation of HBr slows down the rate of the reaction.

2 Only the hydrogen and iodine reaction could be a single step reaction.

3 Doubling the concentration of the halogen in each reaction doubles each rate of reaction.

A 1, 2 and 3 **B** 1 and 2 only **C** 2 and 3 only **D** 1 only